PERTURBATION THEORY WITH BAND-MATRIX PROPAGATORS

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A new version of the Rayleigh–Schrödinger type of perturbation theory is presented. It is based on a rearrangement of hamiltonians $H=T+\lambda V$ containing a non-diagonal (band-matrix, strong-coupling) zero-order component T. Its efficiency is illustrated on the anharmonic oscillator.

The various realistic quantum hamiltonians H do not possess such an exactly solvable approximant H_0 that the corresponding bound-state problem

$$H|\psi\rangle=E|\psi\rangle \tag{1}$$

may be solved perturbatively. Indeed, whenever the perturbation $\lambda H_1=H-H_0$ proves "large", the asymptotic series of the Rayleigh–Schrödinger (RS) type

$$E=E_0+\lambda E_1+...+\lambda^k E_k+\mathrm{O}(\lambda^{k+1}),$$
$$|\psi\rangle=|\psi_0\rangle+\lambda|\psi_1\rangle+...+\lambda^k|\psi_k\rangle+\mathrm{O}(\lambda^{k+1}), \tag{2}$$

do not represent a reasonable approximation scheme at any order k (cf., e.g., ref. [1]).

Considerable attention is currently being paid to an improvement of this situation. The various resummation techniques are often applied to (2) [2]. Here, we shall advocate an alternative approach oriented to an optimization of H_0 itself [3].

First, we notice that the formal background of the explicit RS formulas (2) is rather contradictory: Usually, we try to keep the perturbation $H-H_0$ small and the matrix H_0 diagonal at the same time. Of course, the former requirement could more easily be satisfied with a band matrix,

$$H=T+\lambda V,\quad |\lambda|\ll 1,$$
$$\langle m|T|n\rangle=0,\quad |m-n|>t, \tag{3}$$

where m, $n\geqslant 0$ and T may have $2t+1$ nonzero diagonals, $t>0$. Unfortunately, the modified RS (MRS) decomposition (3) would imply

(i) a numerical character of evaluation of the unperturbed propagators, and/or

(ii) a complete non-diagonality of their matrix form

$$R=Q\frac{1}{E_0-QTQ}Q, \tag{4}$$

where Q is a projector, say, $Q=1-|0\rangle\langle 0|$.

In our preceding papers [4], the MRS shortcoming (i) has been circumvented: for the simplest nontrivial "next-to-RS" choice of $t=1$, all the matrix elements of (4) were shown to be expressible in terms of certain auxiliary analytic continued fractions [5]. Of course, a generalization (3) of the $t=1$ restriction is quite easy. It is sufficient to replace the continued fractions by their $(t\times t)$-dimensional matrix analogues if needed [6].

In the present Letter, we intend to discuss (and get rid of) the general-matrix structure (ii) of R. It doubles unnecessarily the number of the intermediate-state summations and, in effect, introduces a new problem of convergence into the MRS perturbative formalism [4].

Our present proposal may be characterized briefly as a modification and improvement of the MRS formalism of refs. [4] and [6]. Its core lies in a transition $R\to\tilde{R}$ to the band-matrix propagators.

For the sake of definiteness, we may take eq. (4)

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as our starting point, and define our input $\tilde{R}$ as a truncated R,

$$\tilde{R}_{mn}=R_{mn},\quad m,n\geqslant 0,\quad |m-n|\leqslant\tilde{t},$$
$$\tilde{R}_{mk}=0,\quad |m-k|>\tilde{t}. \qquad (5)$$

Then, the difference $\tilde{R}-R$ myst be small – this condition may be expected to restrict our choice of the "trimming parameter", $\tilde{t}\geqslant\tilde{t}_0$.

Of course, the modification $R\to\tilde{R}$ of propagators is equivalent to the modification of their denominators – we shall write

$$\tilde{R}=Q\frac{1}{E_0-Q\tilde{T}Q}Q, \qquad (6)$$

with the tilded $\tilde{T}$ equal to T outside the Q-projected subspace. Thus, it remains for us to interpret the new "unperturbed hamiltonian" $\tilde{T}$ as a small modification of the original T and recall the old MRS formalism [4], listing only its modifications caused by the present tilded rearrangement $H=\tilde{T}+\lambda\tilde{V}$ of eq. (3).

In the first step, we introduce a separable Hartree–Fock-like field again,

$$H=H_0+\lambda H_1,$$
$$H_0=\tilde{T}+\tilde{g}|0\rangle\langle 0|,\quad \lambda H_1=H-H_0. \qquad (7)$$

The unperturbed equation $H_0|\psi_0\rangle=E_0|\psi_0\rangle$ will then read

$$(\tilde{T}+\tilde{g}|0\rangle\langle 0|-E_0)|\psi_0\rangle=0 \qquad (8)$$

and becomes solvable by the formulas

$$|\psi_0\rangle=\text{const}\times|\varphi\rangle,$$
$$|\varphi\rangle=|0\rangle+\tilde{R}T|0\rangle,$$
$$\tilde{g}=\tilde{g}(E_0)=E_0-\langle 0|T|\varphi\rangle. \qquad (9)$$

Obviously, we may treat E_0 as a free parameter, provided only that $\tilde{R}\neq\infty$ and that the auxiliary coupling remains sufficiently small, $\tilde{g}(E_0)=O(\lambda)$.

In the second step, we introduce a normalization requirement

$$|\psi_0\rangle=|\varphi\rangle,\quad \langle 0|\psi_k\rangle=0,\quad k\geqslant 1, \qquad (10)$$

and insert (2) and (10) in (1). This gives the unperturbed eq. (8) accompanied by the recurrent algebraic definition of corrections,

$$(H_0-E_0)|\psi_k\rangle+H_1|\psi_{k-1}\rangle=\sum_{m=1}^{k}|\psi_{k-m}\rangle E_m,$$
$$k=1,2,\dots, \qquad (11)$$

i.e.,

$$E_1=(\langle\varphi|\tilde{V}|\varphi\rangle-\tilde{g}/\lambda)/\langle\varphi|\varphi\rangle,$$
$$|\psi_1\rangle=\tilde{R}(\tilde{V}-E_1)|\varphi\rangle,$$
$$E_2=\langle\varphi|(\tilde{V}-E_1)\tilde{R}(\tilde{V}-E_1)|\varphi\rangle/\langle\varphi|\varphi\rangle, \qquad (12)$$

etc. [4]. It is only necessary to keep in mind here that the tilded perturbation $\tilde{V}$ differs from the original operator V by the matrix elements computed by an inversion of $\tilde{R}$, say, by means of the $(\tilde{t}\times\tilde{t})$-dimensional matrix continued fraction technique again,

$$\lambda\tilde{V}=\lambda V+\lambda(Q\tilde{V}Q-QVQ)$$
$$=\lambda V+QTQ-Q\tilde{T}Q. \qquad (13)$$

Now, our basic assumption may be formulated as the existence of such a cut-off parameter $\tilde{t}<\infty$ in (5) that the difference between V and $\tilde{V}$ remains small.

At present, our assumption may only be tested numerically. For this purpose, we have chosen the simple one-particle example of ref. [4] with the s-wave anharmonic potential $V(r)=r^2+r^4$, harmonic oscillator basis $|n\rangle$ and with the results confirming nicely our present a priori expectations:

(1) In full analogy with the $\tilde{t}=\infty$ conclusions of refs. [4] and [6], the present $\tilde{t}<\infty$ results remain again almost independent of the choice of E_0 from a broad vicinity of the exact value. For the ground state, we have fixed $E_0=5.0$ here. Similarly, the various choices of the "off-diagonality" $\tilde{t}$ in the matrix elements in (3)

$$\langle n|T|n+1\rangle=\langle n+1|T|n\rangle=\gamma\langle n|H|n+1\rangle$$
$$[=\gamma\langle n+1|H|n\rangle=\gamma\beta_n(4n+5),$$
$$\beta_n=(n+1)^{1/2}(n+3/2)^{1/2}],$$
$$\langle n|T|n\rangle=\langle n|H|n\rangle\quad(=6n^2+13n+27/4), \qquad (14)$$

(cf. ref. [4]) lead to the same overall pattern in the results – we have picked $\gamma=0.75$ in what follows, and display the sample of our numerical results in fig. 1

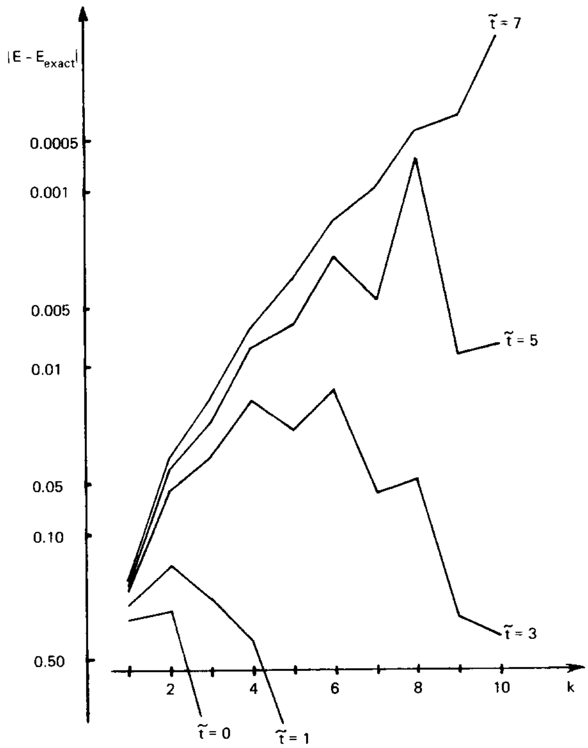

Fig. 1. The precision of energies as a function of the TMRS order k and cut-off $\tilde{t}$.

Table 1
$E-E_{\text{exact}}$. A sample of convergence of the TMRS energies.

TMRS cut-off $\tilde{t}$	TMRS-order k				
	1	2	3	4	5
0	0.3071	−0.2791	1.0850	−3.3104	11.5881
1	0.2476	0.1562	0.2426	0.4268	0.9003
2	0.2197	0.0626	0.0718	−0.0024	0.1629
3	0.2053	0.0568	0.0358	0.0173	0.0255
4	0.1982	0.0474	0.0263	0.0115	0.0113
5	0.1946	0.0416	0.0212	0.0091	0.0061
6	0.1928	0.0384	0.0188	0.0075	0.0043
7	0.1919	0.0366	0.0176	0.0068	0.0035
8	0.1914	0.0357	0.0170	0.0064	0.0032
9	0.1912	0.0351	0.0167	0.0062	0.0030
10	0.1910	0.0348	0.0166	0.0062	0.0030
11	0.1910	0.0347	0.0165	0.0061	0.0030
12	0.1909	0.0346	0.0165	0.0061	0.0030
13	0.1909	0.0345	0.0165	0.0061	0.0030

and table 1. We see that a very good precision is obtainable with quite small $\tilde{t}$ – this is the main conclusion of the whole test.

(2) For all $\tilde{t}<\infty$, an overall MRS convergence pattern seems to preserve an asymptotic-series character (see fig. 1). The precision of energies as well as an optimal MRS order k [7] increase with the increasing cut-off parameter $\tilde{t}$.

(3) For small $\tilde{t}$, table 1 exhibits both the oscillatory and non-oscillatory asymptotic divergence, presumably for the even and odd $\tilde{t}$, respectively. In particular, this divergence is very quick for $\tilde{t}<t$. Its well-known RS example [8] is also reproduced here for diagonal $\tilde{R}$ with $\tilde{t}=0$.

(4) The rate of divergence decreases with increasing $\tilde{t}$. A priori, this may be understood as a consequence of an improving $T\to\tilde{T}$ transfer of information about the strong-coupling part of the hamiltonian.

(5) For $\tilde{t}>t$, the errors induced by the cut-off (5) behave more or less like a geometric series. For example, we may check in table 1 that the MRS first-order deviations $\Delta(\tilde{t})=E-E_{\text{exact}}$ of the ground-state energies from their exact values $E_{\text{exact}}=4.6488127...$ represent only about 4% of the exact value. The $\tilde{t}$-dependent values $\Delta(\tilde{t})$ differ from their $\tilde{t}\to\infty$ limit $\Delta(\infty)$ also by less than 4% (2%, 1%, ...) for $\tilde{t}=4$ ($\tilde{t}=5$, 6, ..., respectively). Such a smoothness of $\Delta(\tilde{t})$ enables us to make a reliable guess of the sufficient truncation $\tilde{t}$ a priori.

(6) The magnitude of the $\tilde{t}$-dependent deviations $\Delta(\tilde{t})-\Delta(\infty)$ is small but it remains almost constant for increasing k. Its order-independence may be understood as a surviving strong-coupling phenomenon related to the magnitude of the modification $Q\tilde{T}Q-QTQ$. Hence, we have to keep more diagonals in $\tilde{R}$ whenever we intend to achieve a significantly higher precision.

We may summarize that our truncation $R\to\tilde{R}$ works well in the anharmonic oscillator example. An additional merit of the whole procedure should also be mentioned in the conclusion: The components

$$\langle m|\varphi\rangle=\langle m|0\rangle+\sum_n\langle m|\tilde{R}|n\rangle\langle n|T|0\rangle \qquad (15)$$

of our zero-order wavefunction satisfy the inequalities $|m-n|\leqslant\tilde{t}$ and $|n-0|\leqslant t$, i.e. they remain nonzero for a limited range of indices only, $m\leqslant t+\tilde{t}$. Hence, due to this finite-dimensional character of

$|\varphi\rangle$'s and to the quasi-diagonal structure of $\tilde{R}$, a number of the intermediate state summations in the tilded MRS (TMRS) formula (12) remains practically the same as in the textbook RS perturbation theory [9]. Thus, our procedure specified by eqs. (3) $(H\to T)$, (4) $(T\to R)$, (5) $(R\to\tilde{R})$, (6) $(\tilde{R}\to\tilde{T})$, (9) $(E_0\to\tilde{g}, |\varphi\rangle)$ and (13) $(\tilde{T}\to\tilde{V})$ reduces significantly the complexity of the original MRS prescription of ref. [4].

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